EEG Spectral Analysis for Inattention Detection in Academic Domain

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***Abstract*—The increasing rate of attention deficit among students, attributed to social media, has far-reaching consequences on their academic performance. Inattention or lack of attention is a state of absent-mindedness or not paying enough attention to the details. A rich body of literature suggests that it is highly associated with underachievement in the academic context. In particular, students who have been clinically diagnosed with Attention Deficit Hyperactivity Disorder(ADHD), a neuro-developmental disorder characterized by inattention, hyperactivity, and impulsivity symptoms are more at risk of under-performance and retention, with some leaving school without a terminal degree. In addition to the academic domain, inattention generally impacts the quality of life and future occupations. Research suggests that ADHD is also attributed to societal and unemployment excess costs as well as productivity loss and healthcare expenses which are estimated to be over $14K per adult in the United States (US). Although more than eight million adults were identified with ADHD in the US by 2018, not all inattention cases are associated with ADHD. Inattention could be caused by several other factors such as stress or anxiety and its early detection and timely intervention is critical, especially in the academic domain. There are existing studies that analyze brain signals by Electroencephalogram (EEG) scans to identify individuals who have ADHD. In this study, we developed a Machine learning (ML) pipeline model that is trained on both ADHD and inattention data to determine if a person is having an attention problem. In the first phase of developing the model, we trained a few different classifiers on a 19-channel public EEG dataset of 60 ADHD and 61 non-ADHD participants. Data analysis showed K-Nearest Neighbor (KNN) classifier outperformed other classifiers with an accuracy of 89%. While many of the existing papers focus on ADHD data, in this work we expand our model to analyze the attention deficiency data as well. To train the model we used an attention dataset which is a collection of 34 recordings of 14-channel EEG that scan the attention states of five young adults. We further implemented Independent Component Analysis (ICA) to reduce noise and dimensions and tried different classifiers to improve the accuracy of the model. We achieved the highest accuracy of 98% with ensemble model classifiers using the improved ML pipeline. To evaluate the proposed model, we conducted a study to record the EEG of 15 young adults while taking a visual reasoning attention test for the duration of 8 minutes. We compared the model output (i.e. binary attention label) and their average test score. Findings showed consistent results between the proposed models' prediction and the visual attention test. this indicates there is a potential to use that test as an alternative to the EEG recording to assess students' attention levels in the academic domain. The goal of this research is not for any diagnosis purposes but merely as a tool to help in the early identification of inattention for timely help and interventions among students.**

***Index Terms*—Attention deficit, Academic performance, Engineering education, EEG, Machine Learning**

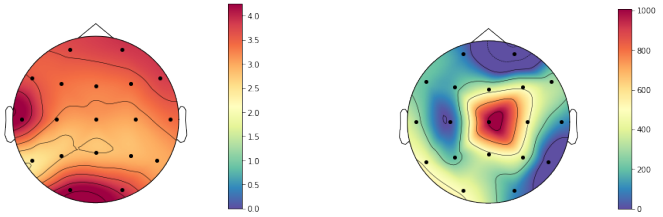
I. Introduction

The increasing rate of attention deficit among students, attributed to social media, has far-reaching consequences on academic performance [2]. Research of 348 undergraduate students from a university in Hong Kong indicates that social media and multitasking were strongly linked to reduced academic performance and that the non-academic use of social media may have a detrimental effect on student's ability to excel academically [2]. Another study of 593 middle school students found that a notable percentage of them began using social media at the age of nine or younger, with 40% accepting friend requests from strangers and having unsupervised social media access [3]. Students who use smartphones absent-mindedly are more susceptible to experiencing instances of inattention and challenges related to attention [4]. Elevated levels of inattention have been associated with a decrease in academic interest, decision-making, self-efficacy, study skills, and GPA [5]. Even more, poor time management and increased course load pressure cause high-level of college drop-out rates [5]. A college dropout rate of 40% among college students in 2022 resulted in a national financial loss of $31 billion [6] [7].

Elevated levels of stress, anxiety, and depression have been found to cause symptoms of inattention [8]. Chronic stress overworks the attention system, making it harder to devote attention [9]. According to the attention control theory, anxiety reduces the effectiveness of the inhibitory function and hinders attention [10]. Attention is crucial in engineering education, as evidenced by the study's findings on ADHD characteristics. Students with ADHD, which encompasses attentional challenges, tend to have lower GPAs in engineering courses compared to their peers. These attentional challenges can lead to difficulties in organization, sustaining effort, and timely task completion among engineering students [25]. A decrease in attention can impair memory, learning, problem-solving, and decision-making, leading to challenges in social interactions, emotional regulation, task completion, and safety [25].

2023 IEEE Frontiers in Education Conference (FIE) | 979-8-3503-3642-9/23/$31.00 ©2023 IEEE | DOI: 10.1109/FIE58773.2023.10343261

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(a) Inattention Brain Scan (b) Attention Brain Scan

Fig. 1: BrainScans

This cascading effect impacts overall cognitive performance and well-being [25]. Most cases involve youngsters, while symptoms such as inattention can often appear later in the adult stage as well. Early detection and timely intervention of inattention issues among students are deemed critical since better attention leads to higher academic performance [11] [12].

II. Methodology

A. Method

Existing research explores the use of EEG scans to identify individuals with ADHD through brain signals. To build upon this, our study presents an ML-based model trained on ADHD and inattention data with the aim of detecting attention problems in students. We used a public EEG dataset to train our developed ML classifier [1]. To develop an ML model, we trained classifiers on a 19-channel public EEG dataset of 60 ADHD and 61 non-ADHD participants [13]. A sample visualization is shown in Fig. 1 displays scans of attention and in-attention states in the brain scan. Our feature extraction process includes the combination of data pre-processing, filtering, dimensionality reduction, and power spectral feature extraction [13] [16]. First, we applied a band-pass filter using the Butterworth design method with a range of 2-40 Hz to reduce noise and artifacts such as cardiac and ocular parameters [15]. Then, we used Independent Component Analysis (ICA) to reduce the dimensions and improve signal quality [15]. Next, we used power spectral density (PSD) to extract features from each channel, analyzing power distribution in frequency bands like alpha and beta waves. Thereafter, we experimented with several standard classifier algorithms from ML libraries such as KNN, SVM, and NN's. We also developed from scratch a Multi-layer Perceptron (MLP) for its high performance [13]. The developed MLP classifier achieved an initial accuracy of 78%.

B. LOSO method

We applied the leave-one-subject-out (LOSO) method [17] that improved the run time and high accuracy of the model with stable results. As we implemented LOSO in each step of data filtering, ICA, and power spectral analysis, the speed of the model improved multi-fold – data processing and training time reduced from hours to minutes. Furthermore, this study implements additional classifiers, such as Decision Tree, Random Forest, XGBoost, and AdaBoost to the ML pipeline, and found that while MLP performed at 88%, other classifiers achieved a stable accuracy rate of 98%.

C. Dataset

While many of the existing papers focus on ADHD data, in this work we expand our model to analyze the attention deficiency data as well. We used two datasets in this study. The ADHD dataset used in this study is available on the IEEE dataport [19]. The dataset consists of EEG data collected from 30 children diagnosed with ADHD and 30 normal children (control group). The second dataset used is sourced from the Kaggle website [20]. The dataset consists of 25-hour 34 EEG recordings collected from 5 young adults who were engaged in attention and inattention tasks designed. The research method to analyze this data involves the following steps as shown in fig. 2:

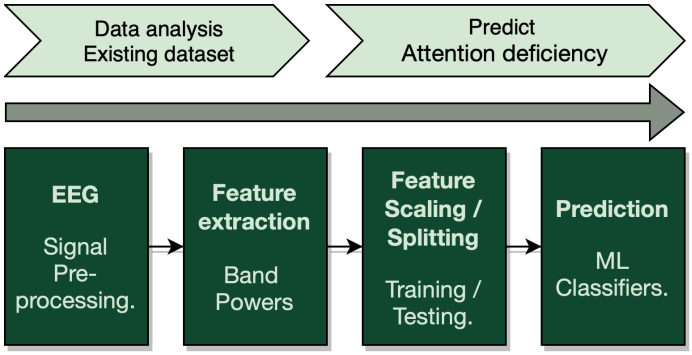


Fig. 2: Method Framework

Metrics

We used TestMyBrain (TMB) Matrix Reasoning and TMB Flanker Attention tests which assess reasoning abilities and attention control, respectively to measure the attention level scores of the participants[18]. The former requires individuals to identify patterns and complete matrices, while the latter measures the ability to focus attention and suppress distracting information during a visual task to produce respective scores.

We also used Muse 2 which is a portable EEG headset that produces raw EEG data from the electrical brain signals (fig. 4) that are captured by four electrodes placed on the head fig. 5 [24]. From this user-friendly wireless device, EEG data can be easily exported for further analysis.

D. Experiment Setup

Potential subjects for the experiment were identified through references from college professors, campus email newsletters, announcements in student groups and associations, and flyers on college notice boards. Fifteen volunteers above 18 years old were selected for this experiment. The experiment involves the following steps as shown in fig. 3.

The Visual Reasoning test [18] contains two parts - TMB Matrix Reasoning and TMB Flanker Attention Test. While the former measures cognitive abilities such as perception, memory, and cognition, the latter measures attention and

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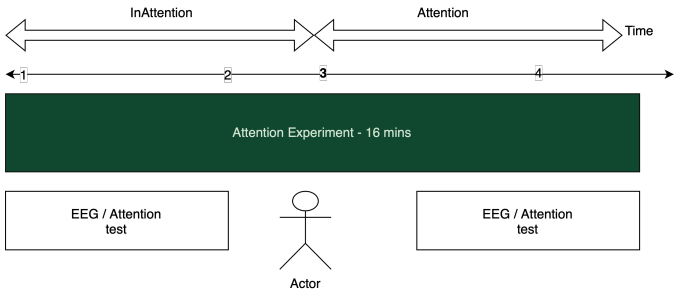


Fig. 3: Method Framework

cognitive control[23]. The participants would take the visual reasoning attention test while wearing the EEG headset (fig. 5).

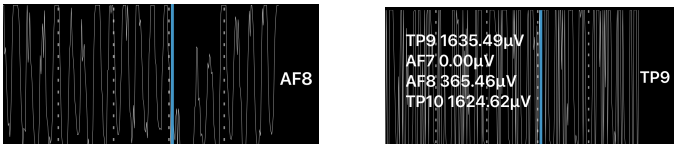


Fig. 4: EEG from Muse 2



Fig. 5: EEG experiment-Museheadset

In the second step, the participant is asked to retake the same visual reasoning attention test online for 8 minutes more. Thereafter, the device is disconnected, and the participant leaves the lab. Thus, in the total experiment time of about 16 minutes, EEG, Attention test scores are recorded for each participant.

We developed the below ML pipeline that performed faster with high performance even on complex large data in our experiments as shown in figure. 6.

The data processing involves several steps, such as data import, data visualization, data cleaning, filtering, and ICA. The power spectral density was then computed for different frequency bands, such as alpha, beta, theta, delta, and gamma. In the detection phase, the data was scaled, label encoded, and split into training, testing, and validation sets as required. The classification was performed using various ML algorithms, and the accuracy of the models was evaluated through validation.

Model evaluation and validation techniques are crucial to ensure the accuracy and reliability of ML models. In our approach, we employed cross-validation to validate our model.

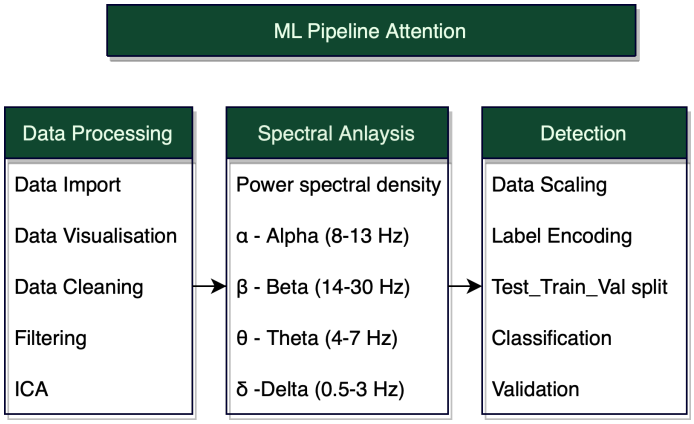


Fig. 6: ML Pipeline

This splits the dataset into several parts and tests the model's performance on each part to obtain an average score, ensuring that the model can generalize well to unseen data. We also verified the effectiveness of the model's detection capabilities using a larger dataset from a different source [22]. The dataset was more extensive than our original one, and the model achieved an accuracy of 95%.

III. Results

From several ML model experiments, we adopted the leave one subject out (LOSO) method over other methods. In this model, a separate dataframe is created with all the participants except for one user. After training the model, a single participant's data is used to test the model. In our experiments, we found that the accuracy of experimented classifiers improved from a worse case of 72% to the best case of 98%. As shown in table I, it was found that ensemble models outperformed the NN model. As mentioned before, this model performance was validated further by processing another larger dataset [22] through the ML pipeline. The ensemble model performed best with an accuracy of 95%. This suggests that the ensemble models such as Random Forest, XGBoost, and AdaBoost could be effective alternatives to NN classifiers in EEG classification. However, in the case of a very large dataset, this could be converse.

TABLE I: Accuracy - Mixed Vs LOSO

Classifier	Mixed	LOSO
XGBoost	0.83	0.98
ADABoost	0.83	0.98
Decision Tree	0.89	0.98
Random Forest	0.83	0.98
MLP Developed	0.78	0.88
Logistic Regression	0.72	0.75

Further enhancement of NN for large data: In addition to the existing features, we included the probability of predictions from the classifiers such as DecisionTree, LightBGM, Logistic Regression, KNN, NaiveBayes, ADABoost, Bagging, RandomForest, SVM and XGBoost to add to the power spectrum dataframe table and used it as input for NNs. This resulted in

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	Reasoning		Shift	Flanker		Shift
	Before	After		Before	After	
1	10	10	0.00%	87.13	87.98	0.97%
2	6	9	33.33%	73.13	73.19	0.08%
3	9	10	10.00%	70.99	68.48	-3.67%
4	6	9	33.33%	63.63	77.48	17.88%
5	7	9	22.22%	73.99	77.48	4.50%
6	6	7	14.29%	70.83	71.98	1.60%
7	8	10	20.00%	67.43	72.53	7.03%
8	6	6	0.00%	72.23	75.1	3.82%
9	8	8	0.00%	69.75	70.78	1.46%
10	9	8	-12.50%	64.36	73.64	12.60%
11	5	6	16.67%	56.55	62.1	8.94%
12	8	8	0.00%	64.1	68.42	6.31%
13	6	6	0.00%	62.34	73.03	14.64%
14	7	8	12.50%	74.82	74.8	0.0%
15	7	8	12.50%	69.96	74.55	6.16%
Mean	7.20	8.13	10.82	69.42	73.44	5.49

TABLE II: Rise in attention

the NNs having access to additional features and larger data, resulting in a significant increase in accuracy(88% to 98%).

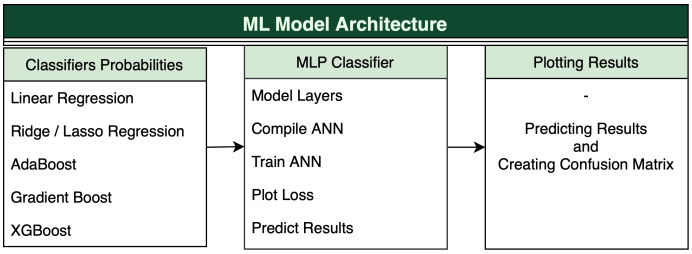
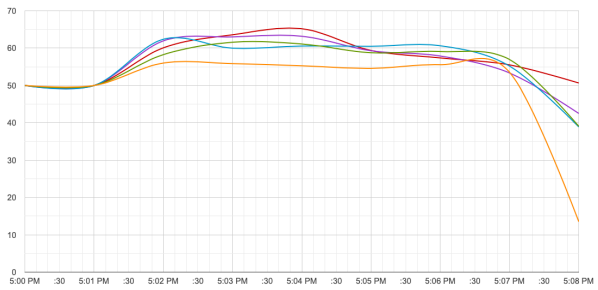


Fig. 7: Enhancing Neural Networks

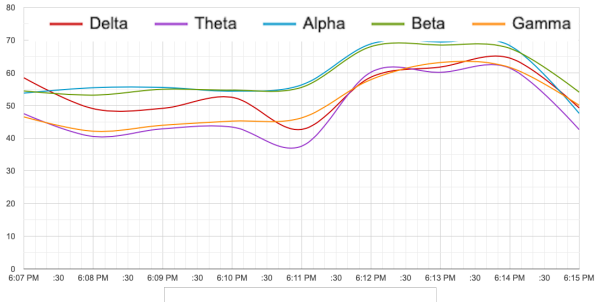
Further to data pre-processing and spectral analysis of the raw EEG data collected, the different band waves as mentioned previously were fed into the trained model. The ensemble and NN models predicted that 11 out of 15 participants were in the 'attention' state during the first test and during the second test, 100% of them moved into the attention state. It can be seen in Figure 8 that alpha and beta waves rise in the graph of the 7th subject indicating a rise in attention. Similar to this, subjects, 2,14, and 15 who were predicted to be in inattention by the model are seen to have started lower than the before mean and moved higher than the after mean. This verifies that the model prediction and the attention test are consistent. However, to generalize this result, repeated experiments with more participants need to be conducted.

Table 2 shows the data of attention tests of the 15 participants.

In summary, ML algorithms, particularly NN classifiers, decision trees, random forests, XGBoost, and AdaBoost have shown promising results for the detection of attention deficiency using EEG data. However, the choice of algorithm depends on the specific characteristics of the dataset and the desired performance metrics. While NN classifiers may be highly efficient, they can be computationally expensive and prone to overfitting, when the data is small and complex. Decision trees and random forests may struggle with handling noisy data or datasets with many features. XGBoost has



(a) EEG Pre-test



(b) EEG post-test

Fig. 8: EEG: Alpha and Beta rise

shown high accuracy, scalability, and fast processing speed, making it well-suited for such complex datasets for real-time applications. But when it comes to large data, NN coupled with ensemble models that we developed may be the best suitable selection.

The results of this study could be useful for the audience in various ways. Firstly, the AI solution we developed could be used in measuring attention by using EEG data from a portable device. This could be useful for educators and researchers who are interested in developing interventions for addressing attention issues in students.

IV. Conclusion and Future Work

The goal of this study was to develop an ML pipeline that predicts the attention level based on the EEG scan. We applied an alternative attention test tool to compare the result of the ML model and the standard attention tool. Early findings show consistency in the predicted attention state by the ML model and the median test score. This research has the potential to help educational researchers to apply such tools to identify the attention states of the students and help them in overcoming the inattention states by appropriate interventions. In future work, we will recruit more participants to test the model and plan to propose interventions such as mindfulness practices that can be applied by educators and students to help them improve their attention levels which leads to better academic achievements.

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